

		$\approx 500 \text{ m/s}$
13	D	Amount of thermal energy supplied to the water $E = mc\Delta\theta + ml$ $= m(c\Delta\theta + l)$ $= 5.00(4190 \times 70 + 2260 \times 10^3)$ $= 1.277 \times 10^7$ $\approx 1.28 \times 10^7 \text{ J}$
14	B	The internal energy of the ideal gas in the canister is independent of the